

Benjamin Pham

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EDUCATION

Virginia Commonwealth University (VCU)

Bachelor of Science: Information Systems

Richmond, VA

August 2021-May 2025

RELEVANT SKILLS

Programming & Scripting: Python, SQL, Java, C#, HTML

AI / ML & Analytics: XGBoost, SHAP, predictive modeling, feature engineering, risk scoring

BI & Visualization: Power BI, dashboard design, data storytelling & insight reporting

Cloud & Platforms: AWS, Microsoft Azure, Windows, Linux, MacOS, Active Directory, DNS

Governance & Compliance: NIST 800-53, MITRE ATT&CK, HIPAA, SEC530

Tools: Splunk Enterprise Security (SPL), CrowdStrike, ServiceNow, Archer, VirusTotal

EXPERIENCE

Virginia Department of Transportation

Security Operations Center (SOC) Analyst

Richmond, VA

December 2025-Present

- Independently triage and analyze 40+ daily cases from large, varied data sources across a statewide environment, exercising ownership and sound judgment to resolve ~95% end-to-end without oversight.
- Query and analyze large-scale, multi-source datasets (DNS, Windows Event Logs, Linux systems, endpoint/network telemetry) using Splunk Enterprise Security (SPL), applying SQL-style querying and correlation to validate data quality and surface anomalies.
- Design and tune custom detection logic and analytics rules in Splunk, building repeatable, rule-based data models that function as a self-service monitoring capability for downstream analysts.
- Apply structured 5 W's (Who, What, When, Where, Why) analysis to drive consistent, well-documented conclusions, translating technical findings into clear insights for non-technical stakeholders.
- Consistently meet strict SLA benchmarks (5-minute detection, 20-minute investigation, 10-minute resolution), reflecting strong prioritization and process-improvement discipline in a fast-paced, metrics-driven environment.
- Align investigative methodology with NIST and MITRE ATT&CK frameworks, applying standards-based rigor to data definitions and analytical decision-making.

Virginia Department of Health

Information Security Analyst

Richmond, VA

July 2024 - July 2025

- Evaluated system documentation and control requirements against SEC530 and NIST 800-53 frameworks, applying data governance and business-definition standards to ensure consistent, well-documented compliance data.
- Authored and maintained governance documentation, including System Security Plans, establishing clear metadata, ownership, and traceability between requirements and implementation to support audit readiness.
- Conducted control assessments, identified data and process gaps, and recommended data-driven remediation strategies, partnering with technical and business stakeholders to strengthen data quality and risk posture.
- Processed and evaluated 1,000+ software installation requests against consistent quality criteria — managing high-volume data review with strong accuracy and attention to detail.

PROJECTS

AI-Powered Credit Card Fraud Detection & Risk Analytics Platform

Personal Project

July 2026

- Built an end-to-end machine learning pipeline in Python and SQL to detect fraudulent credit card transactions across a dataset of 590,000+ records.
- Engineered features and resolved class imbalance to improve model reliability on rare-event (fraud) prediction.
- Trained and validated an XGBoost classification model, evaluating performance via precision, recall, F1-score, and ROC-AUC to optimize detection accuracy.
- Applied SHAP (SHapley Additive Explanations) for model interpretability, surfacing the transaction attributes driving fraud risk to support explainable, data-driven decision-making.
- Designed a risk-scoring framework segmenting transactions into Low/Medium/High risk tiers to streamline investigation prioritization.
- Quantified financial exposure from predicted fraudulent transactions, translating model output into business-facing revenue-risk metrics.